J.J. College of Engineering and Technology

Department of Mechanical Engineering

ME25CO8
Metallurgy and Materials Science

Student Notes
Unit - V: Mechanical Properties and Deformation Mechanisms

Scheduled Hours: 9 Periods

Academic Year : 2026–2027
Year and Semester : 2nd Yr and III Sem

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Course Objectives

1. To learn the construction of phase diagrams and the use of the Iron–Iron carbide phase diagram for understanding microstructure formation.
2. To learn the selection and application of various heat treatment processes and their influence on microstructure formation.
3. To illustrate the different types of ferrous and non-ferrous alloys and their uses in the engineering field.
4. To illustrate different polymers, ceramics and composites and their uses in the engineering field.
5. To learn various testing procedures and failure mechanisms in engineering materials.

Full Course Content

Course Code: ME25CO8 **Course Title:** Metallurgy and Materials Science

Unit	Content
UNIT I	**CONSTITUTION OF ALLOYS AND PHASE DIAGRAMS**: Constitution of alloys - solid solutions, substitutional and interstitial phase diagrams, isomorphous, eutectic, eutectoid, peritectic and peritectoid reactions, Iron–Iron carbide equilibrium diagram. Classification of steel and cast-iron microstructure, properties and application.
UNIT II	**HEAT TREATMENT**: Definition - full annealing, stress relief, recrystallisation and spheroidising - normalizing, hardening and tempering of steel. Isothermal transformation diagrams - cooling curves superimposed on I.T. diagram - continuous cooling transformation (CCT) diagram - austempering, martempering - hardenability, Jominy end quench test - case hardening, carburizing, nitriding, cyaniding, carbonitriding - flame and induction hardening - vacuum and plasma hardening - thermo-mechanical treatments - elementary ideas on sintering.
UNIT III	**FERROUS AND NON-FERROUS METALS**: Effect of alloying additions on steel (Mn, Si, Cr, Mo, Ni, V, Ti and W) - stainless and tool steels - HSLA - maraging steels - grey, white, malleable, spheroidal alloy cast irons, copper and its alloys - brass, bronze and cupronickel - aluminium and its alloys; Al–Cu precipitation strengthening treatment - titanium alloys, Mg-alloys, Ni-based super alloys - shape memory alloys - properties and applications - overview of materials standards.
UNIT IV	**NON-METALLIC MATERIALS**: Polymers - types of polymers, commodity and engineering polymers - properties and applications of PE, PP, PS, PVC, PMMA, PET, PC, PA, ABS, PAI, PPO, PPS, PEEK, PTFE, thermoset polymers, urea and phenol formaldehydes - nylon, engineering ceramics - properties and applications of Al_2O_3, SiC, Si_3N_4, PSZ and SIALON - intermetallics - composites - matrix and reinforcement materials - applications of composites - nano composites.
UNIT V	**MECHANICAL PROPERTIES AND DEFORMATION MECHANISMS**: Mechanisms of plastic deformation, slip and twinning - types of fracture - fracture mechanics - Griffith's theory - testing of materials under tension, compression and shear loads - hardness tests (Brinell, Vickers and Rockwell), micro and nano-hardness tests, impact test Izod and Charpy, fatigue and creep failure mechanisms.

Total: 45 periods (9 periods each)

Course Outcomes

At the end of the course, the students would be able to:

1. Explain alloys and phase diagrams, the Iron–Iron carbon diagram and steel classification.
2. Explain isothermal transformation, continuous cooling diagrams and different heat treatment processes.
3. Clarify the effect of alloying elements on ferrous and non-ferrous metals.
4. Summarize the properties and applications of non-metallic materials.
5. Explain the testing of mechanical properties.

Unit V Learning Focus

For examination preparation, Unit V should be studied with the following sequence:

1. Distinguish elastic and plastic deformation and identify the two important mechanisms: slip and twinning.
2. Understand fracture from the engineering viewpoint: ductile/brittle response, cleavage/shear processes and crack propagation.
3. Understand the purpose of fracture mechanics and the energy basis of Griffith's theory for brittle fracture.
4. Relate tensile, compression and shear tests to the mechanical properties required in components.

5. Compare Brinell, Vickers and Rockwell hardness tests by indenter, measured quantity and application.
6. Understand microhardness and nanoindentation for small regions, coatings, case depth and weld zones.
7. Compare Izod and Charpy impact tests and connect impact energy to sudden loading and notch sensitivity.
8. Explain fatigue as progressive damage under repeated loading and interpret the S–N curve qualitatively.
9. Explain creep as time-dependent deformation under sustained stress at elevated temperature and identify primary, secondary and tertiary creep.

Engineering Context and Application Link

Why this unit matters: A component is not selected only because of its material name. Engineers need to know how the material deforms, how cracks develop, and how it behaves under static, impact, repeated and long-duration high-temperature loading.

Mechanical engineering relevance

- **Automotive:** shafts, crankshafts, gears, springs, fasteners and sheet parts require tensile, hardness, impact and fatigue considerations.
- **Manufacturing:** forging, rolling, drawing and forming rely on controlled plastic deformation; heat treatment is often checked by hardness testing.
- **Power and thermal systems:** boiler tubes, steam pipes, turbine parts and hot fasteners require creep-resistant materials and long-term life assessment.
- **Structural and transport systems:** bridges, axles, welded structures and aircraft parts require fracture, impact and fatigue awareness.
- **Surface engineering:** case-hardened gears, coatings and weld heat-affected zones are evaluated using microhardness or nanoindentation when conventional impressions are too large.

Simple activity/project links

1. Compare a polished and scratched metal surface and discuss why fatigue cracks preferentially initiate at stress concentrators.
2. Prepare a test-selection chart that maps a component (shaft, gear, casting, boiler tube, coating) to the most relevant mechanical test.
3. Use a paper clip or thin wire to demonstrate repeated bending and connect it to fatigue crack initiation and growth.
4. Compare the three creep stages on a schematic curve and identify which stage dominates normal long-term service.

Text Books

1. Kenneth G. Budinski and Michael K. Budinski, *Engineering Materials*, Prentice Hall of India Private Limited, 9th Edition, 2018.
2. Sydney H. Avner, *Introduction to Physical Metallurgy*, McGraw Hill Book Company, 1994.

References

1. A. Alavudeen, N. Venkateshwaran, and J. T. Winowlin Jappes, *A Textbook of Engineering Materials and Metallurgy*, Laxmi Publications, 2006.
2. Amandeep Singh Wadhwa and Harvinder Singh Dhaliwal, *A Textbook of Engineering Material and Metallurgy*, University Sciences Press, 2008.
3. G. S. Upadhyay and Anish Upadhyay, *Materials Science and Engineering*, Viva Books Pvt. Ltd., New Delhi, 2020.
4. V. Raghavan, *Materials Science and Engineering*, Prentice Hall of India Pvt. Ltd., 6th Edition, 2019.
5. William D. Callister, *Materials Science and Engineering*, Wiley India Pvt. Ltd., 2nd Edition Reprint, 2019.
6. User-provided Unit V lecture handout, Griffith theory derivation handout, and Unit V testing/fatigue/creep blackboard notes.

Source alignment note: The supplied Unit V lecture handout is used as the main source for slip, twinning, fracture, tensile/compression testing, impact, hardness, fatigue and creep. The supplied Griffith handout is used for

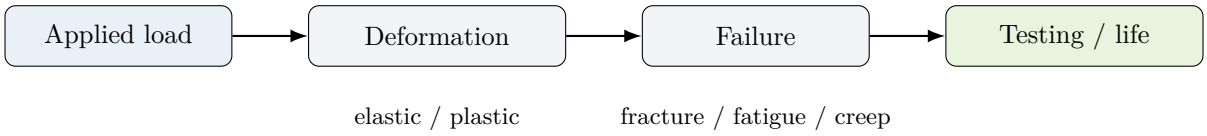
the Griffith energy balance and critical fracture stress. The supplied testing/fatigue/creep blackboard notes are used to present the later syllabus topics in a concise second-year engineering format. The uploaded Unit IV student notes are used as the formatting and organization model for this document.

1 Introduction to Unit V

Mechanical properties describe how a material responds when a load is applied. A component may be pulled, compressed, sheared, struck suddenly, loaded repeatedly or held under load for a long time at high temperature. Each loading condition can produce a different response.

The unit connects three questions:

1. **How does permanent deformation occur?** - slip and twinning.
2. **How and why does fracture occur?** - fracture mechanisms and Griffith's energy approach.
3. **How do engineers measure and predict material behaviour?** - mechanical tests, fatigue and creep.



Exam point: For every topic, connect the loading condition, material response, measurable property and one practical component.

2 Plastic Deformation

2.1 Elastic and Plastic Behaviour

When a material is stressed below its elastic limit, the deformation is temporary and the original dimensions are recovered when the load is removed. When the material is stressed beyond its elastic limit, permanent or plastic deformation occurs.

Plastic deformation is central to mechanical manufacturing because processes such as stamping, pressing, rolling, forging, drawing and extrusion intentionally change the shape of metals.

```
ELASTIC: temporary deformation -> recovers after unloading
PLASTIC: permanent deformation -> does not fully recover
Manufacturing uses controlled plastic deformation.
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2.2 Mechanisms of Plastic Deformation

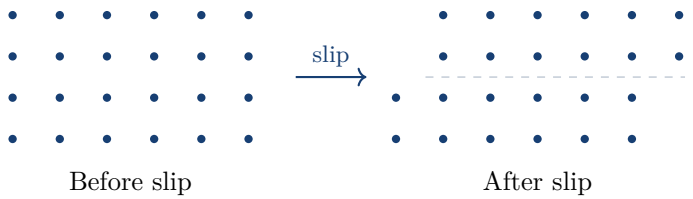
The main mechanisms are:

- **Slip**
- **Twinning**
- In some cases, a combination of slip and twinning.

2.3 Slip

Slip is the relative displacement of one part of a crystal with respect to another along specific crystallographic planes and directions. The supplied lecture notes describe the process using the analogy of cards sliding in a deck. Slip occurs most readily in closely packed directions because less energy is required for atomic movement.

In a simplified picture, atoms move approximately a whole interatomic spacing and the overall lattice orientation is retained. At the surface, slip appears as thin slip lines; surface polishing can remove this visible evidence.



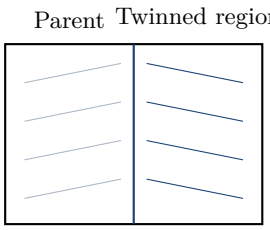
Remember: Slip changes shape mainly by sliding. The lattice orientation is not fundamentally reoriented.

2.4 Twinning

In twinning, atoms move only a fraction of an interatomic spacing in a coordinated manner, causing part of the lattice to change orientation. The twinned region has a characteristic relation to the parent crystal and appears as

a broad band under the microscope. Unlike superficial slip lines, evidence of twinning is not removed simply by polishing the surface.

The supplied source distinguishes deformation/mechanical twins and annealing twins. Mechanical twins are especially associated with some HCP metals, while annealing twins are common in FCC metals after prior working and heat treatment.



Twin boundary

2.5 Slip versus Twinning

Feature	Slip	Twinning
Atomic movement	Approximately whole atomic spacing in the simplified description	Fractional atomic spacing with coordinated movement
Microscopic appearance	Thin slip lines	Wide bands
Lattice orientation	Overall orientation remains essentially unchanged	Orientation changes in the twinned region
Surface polishing	Surface evidence of slip lines may disappear	Twin evidence is not simply removed by polishing
Engineering role	Major plastic-deformation mechanism	Additional deformation/reorientation mechanism

3 Fracture

3.1 Basic Concept

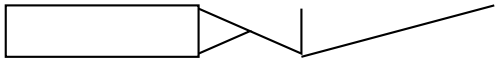
Fracture is the separation of a material such that its load-carrying capacity is lost. At the atomic level fracture requires breaking bonds and creating new crack surfaces. At larger scales, cracks propagate through or along the microstructure until the component separates.

The supplied lecture notes describe cleavage as bond separation mainly normal to the fracture plane and shear fracture as a highly localized plastic-deformation process. Real engineering fractures may contain more than one mechanism.

3.2 Ductile and Brittle Fracture

For second-year engineering use, the most useful first distinction is between ductile and brittle response.

Feature	Ductile fracture	Brittle fracture
Plastic deformation	Significant before final separation	Very small or limited
Typical surface/shape	Necking, fibrous/shear regions, cup-and-cone pattern in tensile bars	Flatter cleavage-type surface may occur
Crack behaviour	Often associated with void formation and coalescence	Cleavage crack can propagate with limited plasticity
Energy absorption	Relatively higher	Relatively lower
Engineering concern	Usually provides deformation before complete failure	Sudden fracture can be critical in structures



Ductile: necking / cup-cone idea



Brittle: little visible necking

3.3 Cleavage and Shear Fracture

Cleavage fracture occurs when concentrated tensile stress at a crack tip separates atomic planes. Under uniaxial tension the crack tends to advance generally perpendicular to the tensile axis.

Shear fracture is associated with localized plastic deformation. In polycrystalline metals, voids may form near inclusions or other stress concentrators, grow and coalesce. In a tensile specimen this mechanism contributes to the familiar cup-and-cone type fracture.

3.4 Transgranular and Intergranular Paths

- **Transgranular fracture:** crack propagates through grains.
- **Intergranular fracture:** crack propagates preferentially along grain boundaries when those boundaries are the weaker path.

3.5 Crack-Tip Loading Modes

The supplied reference identifies three ideal crack-tip separation modes:

- **Mode I - opening mode:** tensile stress acts normal to the crack faces.
- **Mode II - sliding mode:** in-plane shear acts normal to the crack front.
- **Mode III - tearing mode:** anti-plane shear acts parallel to the crack front.

Exam point: Mode I opening is the most direct connection to Griffith's brittle-fracture treatment in this unit.

4 Fracture Mechanics and Griffith's Theory

4.1 Why Fracture Mechanics Is Needed

Real materials contain flaws, notches, pores and cracks. These discontinuities concentrate stress locally, so actual fracture strength can be much lower than the strength expected for an ideal defect-free solid. Fracture mechanics therefore considers both the applied loading and the size/geometry of flaws.

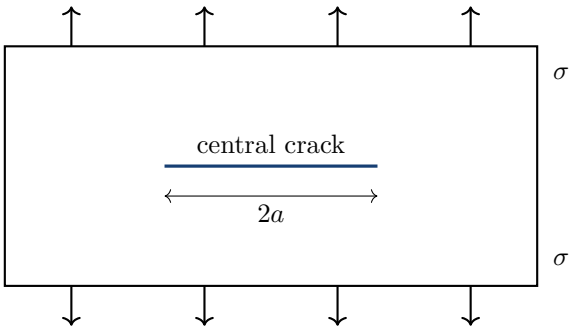
Engineering connection: A small scratch in glass, a machining notch, a weld defect or a sharp keyway can become a local crack-initiation or fracture-critical region.

4.2 Griffith's Central Idea

For an ideally brittle material, crack growth involves two competing energy changes:

1. elastic strain energy is released as the crack extends;
2. energy is required to create two new fracture surfaces.

A crack can grow when the energy released by extension is sufficient to supply the surface-energy requirement.



4.3 Energy Balance

Consider a thin plate of unit thickness containing a central crack of total length $2a$ under tensile stress σ.
The surface energy needed to create the two crack surfaces is

$$U_s = 4a\gamma$$

where γ is surface energy per unit area.
For the ideal thin-plate Griffith model, the elastic strain energy released is

$$U_e = -\frac{\pi\sigma^2 a^2}{E}$$

where E is Young's modulus. The negative sign indicates energy released from the system.
Hence the energy change associated with the crack is

$$U = U_s + U_e = 4a\gamma - \frac{\pi\sigma^2 a^2}{E}.$$

4.4 Critical Condition and Griffith Equation

At the critical condition for crack extension,

$$\frac{dU}{da} = 0.$$

Differentiating,

$$4\gamma - \frac{2\pi\sigma^2 a}{E} = 0.$$

Therefore,

$$\boxed{\sigma_f = \sqrt{\frac{2E\gamma}{\pi a}}}$$

for the ideal brittle, thin-plate form used in this unit.

Remember: The most important physical relation is $\sigma_f \propto 1/\sqrt{a}$. A larger crack requires a lower applied stress for fracture.

4.5 Interpretation and Limitations

- Larger crack size a reduces the fracture stress.
- Greater surface energy γ raises the energy required for new crack surfaces.
- The original Griffith treatment is most directly applicable to ideally brittle behaviour with little plastic deformation.
- In ductile materials, crack-tip plastic deformation consumes additional energy and the simple brittle model is not sufficient by itself.

5 Testing of Engineering Materials

5.1 Why Do We Test Materials?

Mechanical testing converts material behaviour into measurable properties for design, manufacturing control, quality assurance and failure prevention. A material name alone does not tell an engineer whether the component is safe for its actual service condition.

5.2 Property-Test-Component Connection

Question asked by engineer	Typical test	Mechanical example
How much pulling load can it carry?	Tensile test	Tie rod, bolt, cable, sheet metal
How does it behave when squeezed?	Compression test	Block, bearing material, brittle specimen
How much sliding force can it resist?	Shear test	Pin, rivet, bolt joint, key
How resistant is the surface to indentation/wear?	Hardness test	Gear tooth, shaft journal, heat-treated surface
How much energy can it absorb under sudden loading?	Impact test	Axle, structural steel, weld, machine/automotive part
How long can it survive repeated loading?	Fatigue behaviour	Rotating shaft, spring, gear, crankshaft
Will it slowly deform at high temperature?	Creep behaviour	Turbine blade, boiler tube, steam pipe

6 Tensile Test

6.1 Purpose and Principle

A standard specimen is pulled axially in a Universal Testing Machine (UTM) until fracture. Load and extension are recorded. The tensile test is one of the most fundamental methods for obtaining mechanical properties.

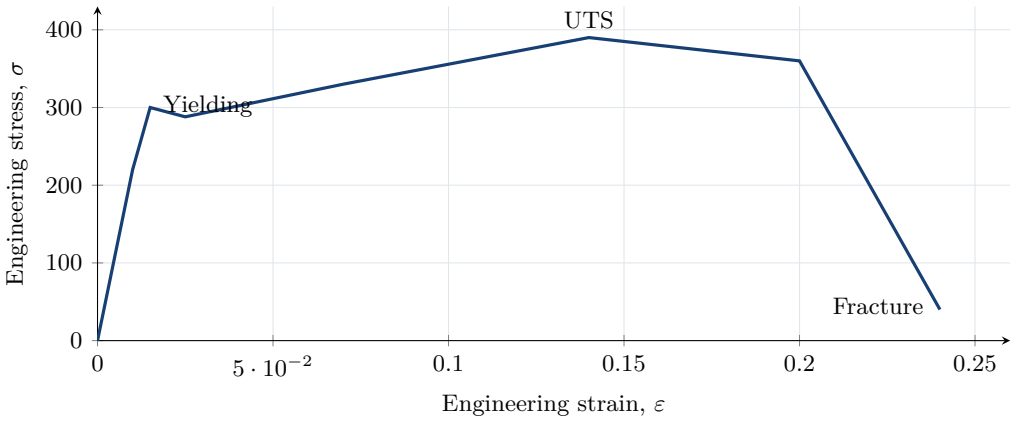
Engineering stress and strain are

$$\sigma = \frac{P}{A_0}, \qquad \varepsilon = \frac{\Delta L}{L_0}$$

where P is applied load, A_0 original cross-sectional area, L_0 original gauge length and ΔL change in gauge length.

6.2 Stress-Strain Behaviour

In the initial linear elastic region, stress is proportional to strain and the slope is Young's modulus E. Beyond the elastic/proportional region, permanent deformation begins. The tensile profile is used to identify yield strength/proof stress, ultimate tensile strength (UTS), fracture and ductility.



6.3 Main Information Obtained

- Young's modulus (stiffness)
- Yield strength / proof stress
- Ultimate tensile strength
- Fracture strength
- Percentage elongation and reduction in area (ductility)

6.4 Simple Test Sequence

1. Measure original specimen dimensions.

2. Grip and align the specimen axially in the UTM.
3. Apply tensile load gradually.
4. Record load and extension.
5. Continue through yielding, plastic deformation, necking and fracture.
6. Construct the stress-strain curve and calculate required properties.

Engineering connection: Bolts and tie rods require adequate yield/tensile strength; sheet metals require sufficient ductility for forming; heat-treated steels are often compared before and after treatment using tensile properties.

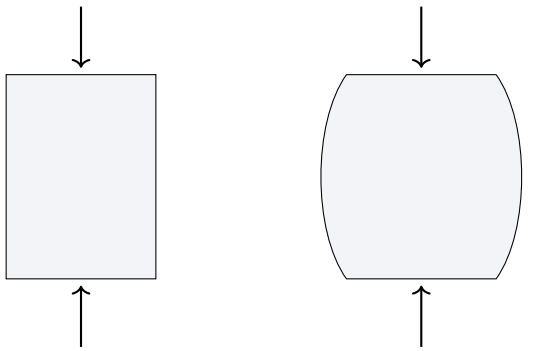
7 Compression Test

7.1 Principle

A specimen is subjected to gradually increasing compressive load. The test measures resistance to shortening, crushing and compressive deformation.

$$\sigma_c = \frac{P}{A_0}, \qquad \varepsilon_c = \frac{\Delta L}{L_0}$$

In compression the specimen shortens and normally expands laterally. Friction between specimen ends and platens can restrict lateral flow and produce barreling in ductile materials.



7.2 Applications

- Cast iron, ceramics and other brittle/low-ductility materials.
- Bearing blocks and compression members.
- Forging and metal-flow studies.
- Cellular/foam materials where crushing and energy absorption are important.

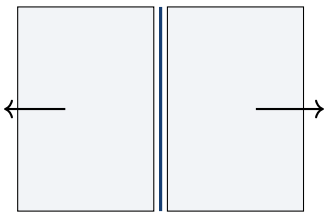
8 Shear Test

8.1 Principle

Shear loading tends to make one part of a material slide relative to the adjacent part. Average shear stress is

$$\tau = \frac{P}{A}.$$

For double shear, two shear planes resist the applied load, so the total resisting area is doubled.



Shear plane

8.2 Applications

Riveted and bolted joints, pins in clevis joints, keys transmitting torque between shaft and hub, and sheet-metal punching/shearing operations.

9 Hardness Testing

9.1 Meaning of Hardness

Hardness is resistance to localized plastic deformation such as indentation, scratching or abrasion. Indentation hardness tests are widely used because they are comparatively quick and require only a small local area.

Remember: Hardness testing is useful for checking heat treatment, wear-resistant surfaces and production consistency, but it is not a substitute for every other mechanical test.

9.2 Brinell Hardness Test

A hard spherical ball is pressed into the surface under a known load. The diameter of the indentation is measured. A larger indentation generally indicates a softer material.

For ball diameter D, indentation diameter d and applied load P,

$$\mathrm{BHN} = \frac{2P}{\pi D\left(D - \sqrt{D^2 - d^2}\right)}.$$

Usefulness:

- large indentation averages local microstructural variations;
- useful for castings and forgings with comparatively coarse/non-uniform structures;
- not preferred for very thin parts or where a large impression is unacceptable.

9.3 Vickers Hardness Test

A square-based diamond pyramid is pressed into the surface. The two diagonals of the square indentation are measured and averaged.

$$HV \approx \frac{1.854P}{d^2}$$

where P is the applied load and d is the mean diagonal length.

Vickers testing is useful for hard materials, small test areas, case-hardened layers, weld regions and microhardness work.

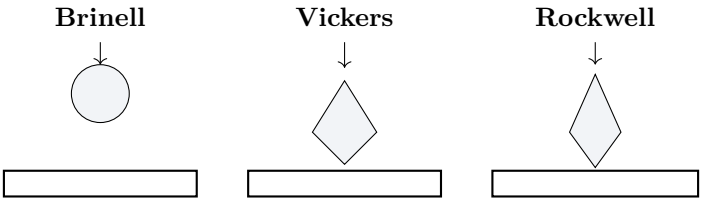
9.4 Rockwell Hardness Test

Rockwell hardness is determined from the permanent depth of indentation produced by a specified minor load followed by a major load. Because the machine gives a direct reading, the test is convenient for production inspection.

Different Rockwell scales use different indenters and loads. A diamond cone is used for harder materials in common scales such as HRC, while ball indenters are used for softer materials.

9.5 Brinell-Vickers-Rockwell Comparison

Feature	Brinell	Vickers	Rockwell
Indenter	Ball	Diamond pyramid	Diamond cone or ball
Measured quantity	Indent diameter	Indent diagonal	Indent depth
Reading	Optical measurement	Optical measurement	Direct machine reading
Good example	Castings / forgings	Hard or small regions	Production components
Main caution	Large impression	Careful optical measurement	Correct scale, thickness and surface required



10 Microhardness and Nanohardness

10.1 Microhardness

Microhardness uses small loads and very small indentations to measure hardness in tiny regions. Vickers or Knoop geometries are commonly used.

Typical applications include:

- case-hardened gear: hardness variation from hard surface toward softer core;
- weld: comparison of weld metal, heat-affected zone (HAZ) and base metal;
- multiphase alloy: local hardness of different microstructural regions;
- thin sheet or small component where a conventional large indentation is unsuitable.

10.2 Nanohardness / Nanoindentation

Nanoindentation uses extremely small loads and penetration depths. Load and penetration depth are continuously recorded so that hardness and indentation modulus can be estimated at very small scales.

Applications include thin protective coatings on cutting tools, MEMS/micro-components, very small phases, and surface-modified layers where a conventional indentation would extend into the substrate.

11 Impact Testing

11.1 Why Impact Testing?

A material that performs well under slowly applied load may behave differently when the load is applied suddenly. Impact tests measure the energy absorbed during rapid fracture of a notched specimen.

$$\text{Impact energy} = \text{initial pendulum energy} - \text{remaining pendulum energy}.$$

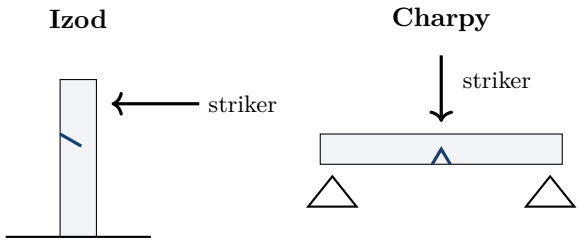
The notch provides a controlled stress concentration so that materials can be compared under rapid fracture conditions.

11.2 Izod Impact Test

- specimen is vertical;
- one end is fixed (cantilever);
- notch faces the striker;
- pendulum strikes the upper part of the specimen;
- energy absorbed in fracture is measured.

11.3 Charpy Impact Test

- specimen is horizontal;
- simply supported at both ends;
- notch is opposite the striker;
- pendulum strikes the centre of the specimen;
- energy absorbed in fracture is measured.



11.4 Izod-Charpy Comparison

Feature	Izod	Charpy
Specimen position	Vertical	Horizontal
Support	Cantilever; one end fixed	Simply supported at both ends
Notch relative to striker	Faces striker	Opposite striker
Typical classroom connection	Comparative impact response of metals/plastics	Structural metals, welds and temperature effects

Engineering connection: Impact testing is relevant when machine/automotive parts may experience shock loads and when structural steels or welds must resist rapid fracture, particularly where notch sensitivity or low-temperature embrittlement is a concern.

12 Fatigue Failure Mechanism

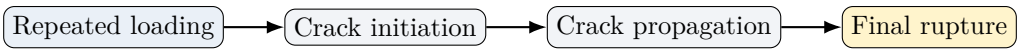
12.1 Definition and Importance

Fatigue is progressive failure caused by repeated or fluctuating stresses. Fatigue fracture can occur even when the maximum applied stress is below the strength required to cause immediate static failure.

A useful classroom question is: *Why can a rotating shaft fail after millions of revolutions even though it never experienced one extreme overload?* The answer is that repetition allows small local damage and cracks to accumulate.

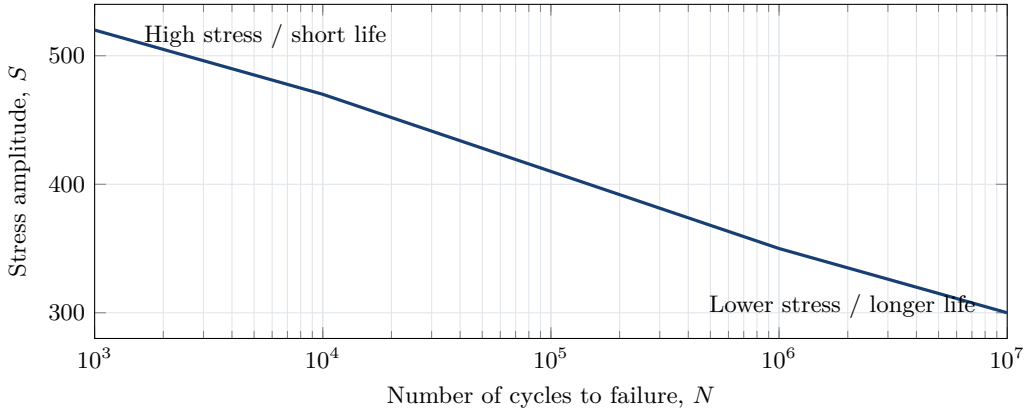
12.2 Three Stages of Fatigue Fracture

1. **Crack initiation:** local irreversible changes occur, often at the surface or at a stress concentrator such as a notch, keyway, thread, scratch or inclusion.
2. **Crack propagation:** the microcrack grows progressively over many load cycles, commonly changing direction so that the main crack advances under the tensile part of the cycle.
3. **Final rupture:** the remaining uncracked section becomes too small to carry the load and fails by final overload.



12.3 S-N Curve

The S–N curve relates cyclic stress level S to the number of cycles to failure N. In general, higher cyclic stress produces shorter fatigue life.



Some ferrous alloys show an approximate endurance limit under specified test conditions. Many non-ferrous alloys do not show a sharp endurance limit, so fatigue strength is stated at a selected number of cycles.

12.4 Factors Affecting Fatigue

- stress concentration: sharp corners, holes, keyways and threads;
- surface finish: scratches and rough surfaces assist crack initiation;
- size and geometry of component;
- residual stress: compressive surface residual stress can improve fatigue resistance, while tensile residual stress may be harmful;
- corrosion/environment;
- mean stress and stress amplitude of the loading cycle.

12.5 Applications

Rotating shafts and axles, crankshafts, gear teeth, coil and leaf springs, threaded fasteners, aircraft structures and bridges all experience repeated service loading.

13 Creep Failure Mechanism

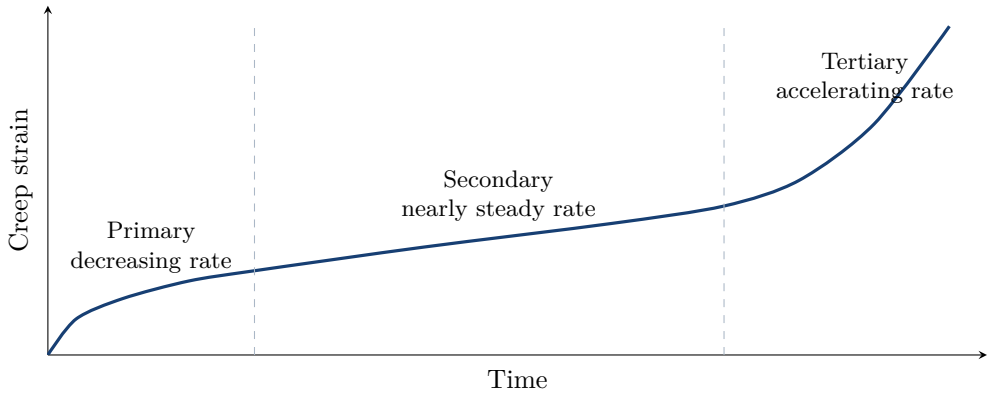
13.1 Definition

Creep is time-dependent permanent deformation under sustained stress, especially at elevated temperature. A component can continue to deform slowly even when the applied load remains essentially constant.

A useful rule of thumb in the supplied teaching notes is that creep becomes increasingly important for many metals above roughly $0.4T_m$, where T_m is the absolute melting temperature in kelvin. The exact temperature depends on material, stress and required service life.

13.2 Creep Curve and Stages

A creep test records strain (or extension) with time under maintained load/stress and temperature.



Stage	Creep rate	Physical interpretation
Primary creep	Decreasing	Initial deformation; strain hardening tends to reduce the rate.
Secondary creep	Approximately steady	Hardening and recovery are roughly balanced; often the longest and most important service stage.
Tertiary creep	Increasing rapidly	Damage, voids, necking or loss of load-bearing area accelerates deformation until rupture.

13.3 Creep Damage and Microstructural Features

The supplied lecture notes describe grain-boundary sliding as an important creep deformation process. During long-term high-temperature exposure, voids can form at grain-boundary junctions or other weak locations, grow and link to form cracks. The effective load-bearing area is gradually reduced until final rupture occurs.

13.4 Mechanical Engineering Applications

- gas-turbine blades: high temperature plus centrifugal stress for long periods;

- boiler superheater/reheater tubes and steam pipes: high temperature plus internal pressure;
- power-plant components requiring long design lives;
- engine exhaust valves/manifolds and other continuously hot components;
- high-temperature fasteners and joints where long-term deformation/relaxation can reduce clamping force.

13.5 Reducing Creep Problems

Use creep-resistant alloys and stable microstructures, reduce operating stress where possible, control temperature, avoid local overheating/stress concentration, specify service life, and inspect critical high-temperature components.

14 Fatigue and Creep - Do Not Confuse Them

Aspect	Fatigue	Creep
Main cause	Repeated / cyclic stress	Sustained stress + time + elevated temperature
Typical example	Rotating shaft, gear, spring	Turbine blade, boiler tube, steam pipe
Damage evolution	Crack initiation → growth → final fracture	Progressive strain → damage → creep rupture
Key plot	S–N curve	Creep strain versus time
Can occur below static yield strength?	Yes, under cyclic loading	Yes, especially at sufficiently high temperature and long time
Design concern	Number and severity of cycles	Stress, temperature and service duration

15 Test Selection - Quick Engineering Guide

Engineering requirement	**Suggested test / concept**	**Example**
Measure stiffness, yield and UTS	Tensile test	Bolt, rod, structural steel
Assess crushing/compressive behaviour	Compression test	Cast iron, block, ceramic
Check pin/rivet/key resistance	Shear test	Clevis pin, riveted joint
Bulk hardness of coarse casting/forging	Brinell	Cast iron / forging
Hardness of small hard region	Vickers	Case-hardened layer, weld
Fast production hardness check	Rockwell	Heat-treated steel parts
Hardness variation across tiny zone	Microhardness	Gear case depth, HAZ
Thin coating / very small phase	Nanoindentation	Tool coating, micro-component
Sudden load / notch toughness	Izod / Charpy	Axle, weld, structural metal
Repeated service loading	Fatigue / S–N behaviour	Shaft, spring, gear
Long-time hot loading	Creep test / creep curve	Turbine blade, boiler tube

16 Important Exam-Oriented Points

16.1 Very Short Answer Points

1. Plastic deformation is permanent deformation remaining after unloading.
2. Slip is relative sliding of crystal regions along preferred planes/directions.
3. Twinning produces coordinated fractional atomic movement and lattice reorientation.
4. Slip lines appear thin; twins appear as broader bands in the supplied reference description.
5. Fracture is the separation of material with loss of load-carrying capacity.
6. Cleavage is associated with bond separation under tensile stress; shear fracture involves localized plastic deformation.
7. Transgranular fracture passes through grains; intergranular fracture follows grain boundaries.
8. Griffith theory balances elastic-energy release against surface-energy creation.

9. For the ideal Griffith model, $\sigma_f \propto 1/\sqrt{a}$.
10. Tensile stress is $\sigma = P/A_0$; engineering strain is $\varepsilon = \Delta L/L_0$.
11. Tensile testing provides E, yield strength, UTS, fracture and ductility information.
12. Compression testing measures behaviour under crushing/shortening loads.
13. Shear stress is commonly written as $\tau = P/A$ for average direct shear.
14. Brinell uses a ball and measures indentation diameter.
15. Vickers uses a diamond pyramid and measures indentation diagonals.
16. Rockwell uses indentation depth and gives a direct hardness reading.
17. Microhardness is useful for small regions such as case depth and weld HAZ.
18. Nanoindentation uses very small loads/depths and records load-penetration response.
19. Izod: vertical cantilever, notch facing striker.
20. Charpy: horizontal simply supported specimen, notch opposite striker.
21. Fatigue develops under repeated/fluctuating loading through initiation, propagation and final rupture.
22. S–N curve relates stress level to cycles to failure.
23. Creep is time-dependent deformation under sustained load, especially at high temperature.
24. Creep has primary, secondary and tertiary stages.

16.2 Important Two-Mark Questions

1. Define plastic deformation. Name its two principal mechanisms.
2. Define slip and slip plane/direction in simple terms.
3. Define twinning and state one difference from slip.
4. Differentiate slip and twinning.
5. Define fracture. What is meant by cleavage fracture?
6. Differentiate transgranular and intergranular fracture.
7. Name the three crack-tip loading modes.
8. State Griffith's basic energy concept.
9. Write the Griffith fracture-stress equation and define the symbols.
10. State the effect of crack length on Griffith fracture stress.
11. Define engineering stress and engineering strain.
12. List four properties obtained from a tensile test.
13. What is barreling in compression testing?
14. Write the expression for average shear stress.
15. Define hardness.
16. State the indenter and measured quantity in Brinell, Vickers and Rockwell tests.
17. What is microhardness? Give two applications.
18. What is nanoindentation? Give one application.
19. Differentiate Izod and Charpy tests.
20. Define fatigue and list the three stages of fatigue fracture.
21. What is an S–N curve?
22. Define creep and list its three stages.
23. Give two high-temperature applications where creep is important.

16.3 Important Five-Mark Questions

1. Explain plastic deformation by slip with a neat sketch.
2. Explain twinning and compare it with slip.
3. Explain ductile and brittle fracture with engineering significance.
4. Explain cleavage, shear, transgranular and intergranular fracture.
5. Explain Griffith's energy balance and the significance of crack size.
6. Explain the tensile test and the engineering stress-strain curve.
7. Explain compression testing and the barreling effect.
8. Explain the principle and applications of Brinell hardness testing.
9. Explain Vickers and Rockwell hardness tests.
10. Compare Brinell, Vickers and Rockwell tests.
11. Explain microhardness and nanoindentation with applications.
12. Compare Izod and Charpy impact tests.
13. Explain the stages of fatigue fracture and factors affecting fatigue life.
14. Explain the S–N curve and its engineering significance.
15. Explain the three stages of creep and the typical creep curve.

16.4 Important Long-Answer Questions

1. Explain the mechanisms of plastic deformation and compare slip and twinning with neat sketches.
2. Discuss fracture mechanisms and classify fracture paths/modes with suitable explanations.
3. Explain Griffith's theory of brittle fracture and derive the critical fracture-stress relation from the energy balance.
4. Explain tensile, compression and shear testing and state the important properties obtained from each.
5. Explain Brinell, Vickers and Rockwell hardness tests and compare their principles, measurements and applications.
6. Explain microhardness, nanoindentation, Izod and Charpy tests with engineering applications.
7. Explain fatigue failure mechanism, S–N behaviour, influencing factors and common mechanical applications.
8. Explain creep failure mechanism, creep curve, microstructural damage and high-temperature applications.

17 Unit Summary

- Plastic deformation is permanent deformation. Slip and twinning are the principal mechanisms considered in this unit.
- Slip is mainly sliding along preferred crystal planes/directions, while twinning reorients a region of the lattice by coordinated atomic movement.
- Fracture involves crack formation/propagation and loss of load-carrying capacity. Cleavage and shear are fundamental fracture processes; cracks may travel through grains or along grain boundaries.
- Fracture mechanics recognizes the importance of flaws. Griffith's brittle-fracture theory compares elastic strain-energy release with the energy needed to create new surfaces.
- Griffith's ideal relation shows that larger cracks reduce the fracture stress: $\sigma_f \propto 1/\sqrt{a}$.
- Tensile testing provides stiffness, yield, UTS and ductility information; compression and shear tests address different loading modes.
- Hardness measures resistance to local indentation/plastic deformation. Brinell measures a ball indentation diameter, Vickers a diamond-pyramid diagonal, and Rockwell indentation depth.
- Microhardness and nanoindentation extend hardness measurement to small regions, thin layers and coatings.
- Izod and Charpy compare energy absorbed during rapid fracture of notched specimens using different specimen support arrangements.

- Fatigue is progressive failure under repeated loading: initiation, propagation and final rupture. The S–N curve links cyclic stress and fatigue life.
- Creep is time-dependent deformation under sustained stress at elevated temperature. Primary, secondary and tertiary creep describe decreasing, nearly steady and accelerating creep rate respectively.
- Mechanical testing must match the service condition; static strength alone does not guarantee safety under impact, cyclic loading or long-term high-temperature service.

18 Glossary of Important Terminologies

The following short definitions are useful for quick revision and two-mark answers.

Term	Short definition / exam point
Elastic deformation	Temporary deformation that is recovered after removal of load, within the elastic range.
Plastic deformation	Permanent deformation that remains after unloading.
Slip	Relative displacement of crystal regions along preferred crystallographic planes and directions.
Slip plane	Crystallographic plane along which slip occurs.
Slip direction	Direction in the slip plane along which atomic displacement occurs.
Twinning	Coordinated atomic movement that reorients a region of the crystal to form a twin.
Twin boundary	Interface separating the parent and twinned regions.
Fracture	Separation of a material leading to loss of load-carrying capacity.
Cleavage	Crack propagation by separation of atomic bonds, commonly associated with brittle fracture.
Shear fracture	Fracture involving intense localized shear/plastic deformation.
Transgranular fracture	Crack propagation through grains.
Intergranular fracture	Crack propagation along grain boundaries.
Mode I	Opening mode; tensile loading normal to crack faces.
Mode II	In-plane sliding mode at the crack tip.
Mode III	Anti-plane tearing/shear mode at the crack tip.
Surface energy, γ	Energy required to create unit new surface area in the Griffith model.
Griffith criterion	Energy balance for crack growth in an ideal brittle material.
Engineering stress	Applied load divided by original cross-sectional area, P/A_0.
Engineering strain	Change in gauge length divided by original gauge length, $\Delta L/L_0$.
Young's modulus, E	Slope of the linear elastic stress-strain region; measure of stiffness.
Yield strength	Stress at which significant plastic deformation begins.
Ultimate tensile strength	Maximum engineering tensile stress reached in the test.
Ductility	Ability to undergo plastic deformation before fracture; often represented by elongation/reduction in area.
Barreling	Bulging of a compressed ductile specimen due mainly to friction at the platens.
Shear stress, τ	Tangential load divided by resisting shear area for average direct shear.
Hardness	Resistance to localized plastic deformation such as indentation.
Brinell hardness	Hardness based on the size of an impression produced by a ball indenter.
Vickers hardness	Hardness based on the diagonal of a square-pyramidal diamond indentation.
Rockwell hardness	Hardness determined from permanent indentation depth under specified minor/major loads.
Microhardness	Small-load indentation hardness measurement for tiny regions, often using Vickers or Knoop geometry.
Nanoindentation	Instrumented very-small-scale indentation recording load and penetration depth.
Impact energy	Energy absorbed by a specimen during rapid fracture in a pendulum impact test.
Izod test	Pendulum impact test using a vertical cantilever specimen with notch facing the striker.
Charpy test	Pendulum impact test using a horizontal simply supported notched specimen.
Fatigue	Progressive damage/failure under repeated or fluctuating stress.
Fatigue crack initiation	First stage in which a small crack forms, often at a surface/stress concentrator.
Fatigue crack propagation	Progressive crack growth over many loading cycles.
S–N curve	Plot relating cyclic stress level to number of cycles to failure.

Term	Short definition / exam point
Endurance limit	Approximate stress level below which some ferrous alloys can show very long fatigue life under specified conditions.
Creep	Time-dependent permanent deformation under sustained stress, especially at elevated temperature.
Primary creep	Creep stage with decreasing creep rate.
Secondary creep	Creep stage with approximately steady/minimum creep rate.
Tertiary creep	Creep stage with accelerating rate leading to rupture.
Stress rupture	Final failure after long-term loading at elevated temperature; closely associated with creep failure.

End of Unit V Student Notes

JJCET Mechanical Engineering Lecture Material